

Second order system :-

8.5 Step Response of a Series RLC Circuit

As we learned in the preceding chapter, the step response is obtained by the sudden application of a dc source. Consider the series RLC circuit shown in Fig. 8.18. Applying KVL around the loop for $t > 0$,

$$L \frac{di}{dt} + Ri + v_c = V_s \quad (8.39)$$

But

$$i = C \frac{dv_c}{dt}$$

Figure 8.17

For Practice Prob. 8.6.

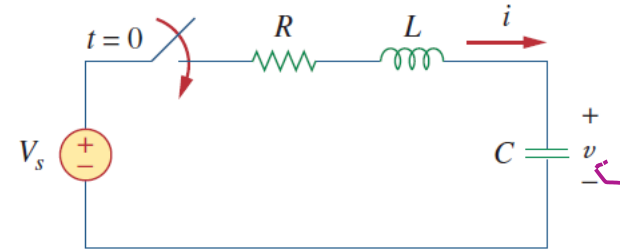


Figure 8.18

Step voltage applied to a series RLC circuit.

$$\therefore L \frac{d}{dt} \left(C \frac{dv_c}{dt} \right) + R C \frac{dv_c}{dt} + v_c = V_s$$

$$\Rightarrow LC \frac{d^2 v_c}{dt^2} + RC \frac{dv_c}{dt} + v_c = V_s$$

$$\Rightarrow \frac{d^2 v_c}{dt^2} + \frac{R}{L} \frac{dv_c}{dt} + \frac{v_c}{LC} = \frac{V_s}{LC}$$

Now, Transient condition! —

$$\frac{d^2 V_c}{dt^2} + \frac{R}{L} \frac{dV_c}{dt} + \frac{V_c}{LC} = 0$$

Let, $V_c = A e^{st}$

Now,
$$\frac{d^2}{dt^2} (A e^{st}) + \frac{R}{L} \frac{d}{dt} (A e^{st}) + \frac{1}{LC} (A e^{st}) = 0$$

$$\Rightarrow s^2 A e^{st} + \frac{R}{L} s A e^{st} + \frac{1}{LC} A e^{st} = 0$$

$$\Rightarrow s^2 + \frac{R}{L} s + \frac{1}{LC} = 0$$

$$\text{So, } s_1 = \frac{-\frac{R}{L} + \sqrt{\left(\frac{R}{L}\right)^2 - 4 \cdot \frac{1}{LC}}}{2}$$

$$s_1 = \frac{-\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}}{2}$$

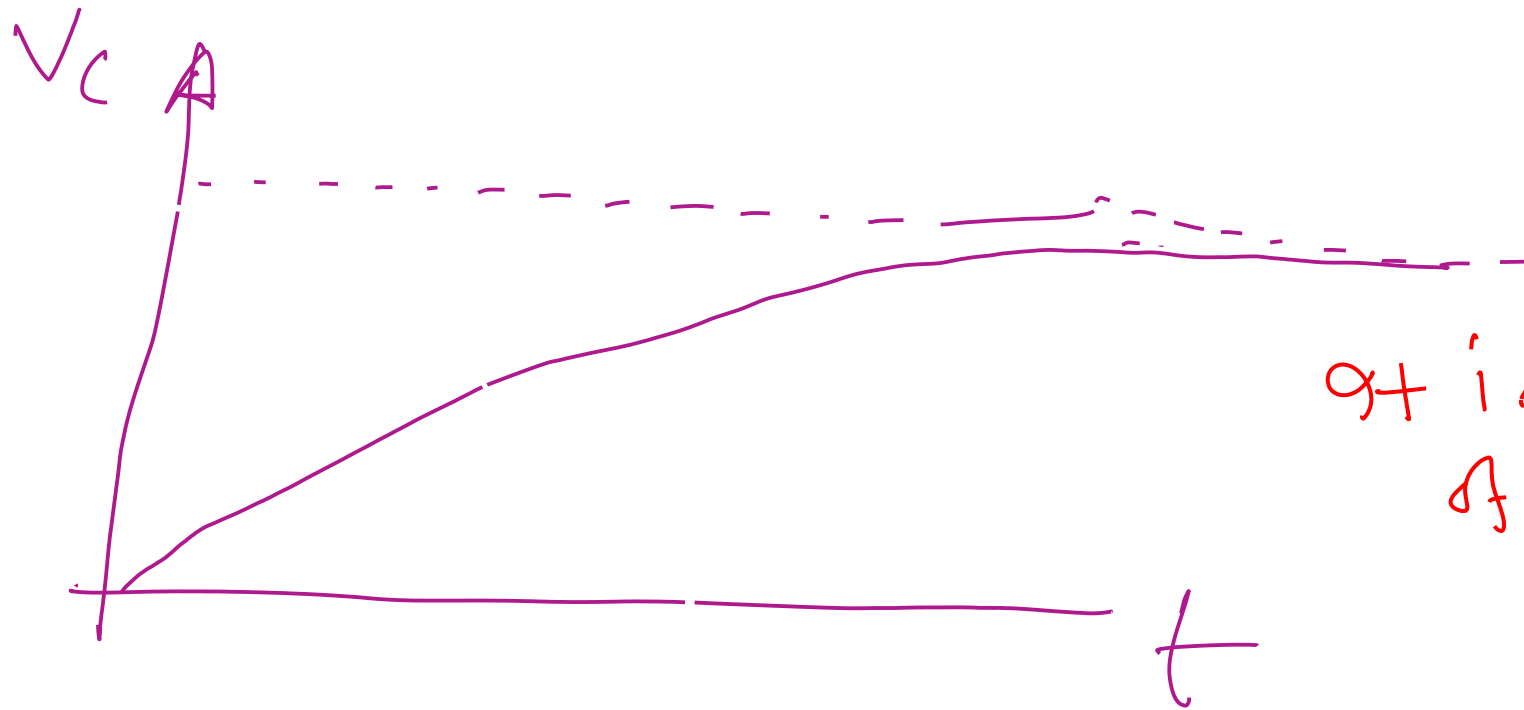
Similarly,

$$s_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}.$$

Now, if $\left(\frac{R}{2L}\right)^2 > \frac{1}{LC}$; roots are real and unequal. So, the response is -

$$V_{C_r}(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \left/ \begin{array}{l} s_1 \text{ and } s_2 \\ \text{roots of the} \\ \text{equation.} \end{array} \right.$$

This is called overdamped case.

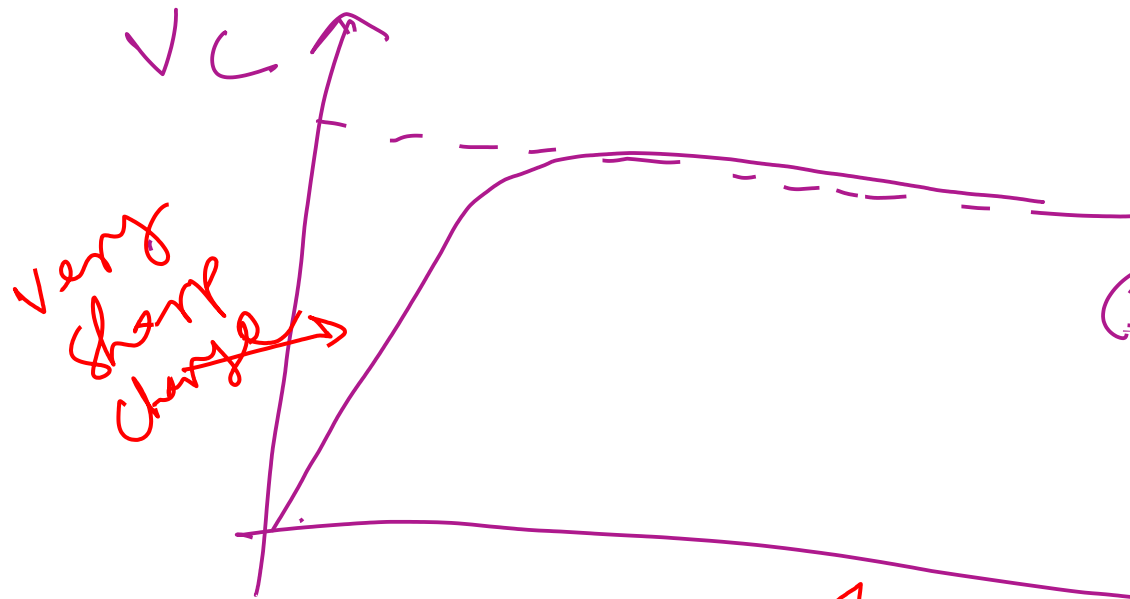


It is the response
of the overdamped
case.

Overdamped case.

Now, when $\left(\frac{R}{2L}\right)^2 = \frac{1}{LC}$, then the roots are real and equal.

so, $V_C(t) = A_1 e^{st} + \underline{t} A_2 e^{st}$



It is called
Critical damped
Case.

Fig. Response of
~~under~~ critical damped case.

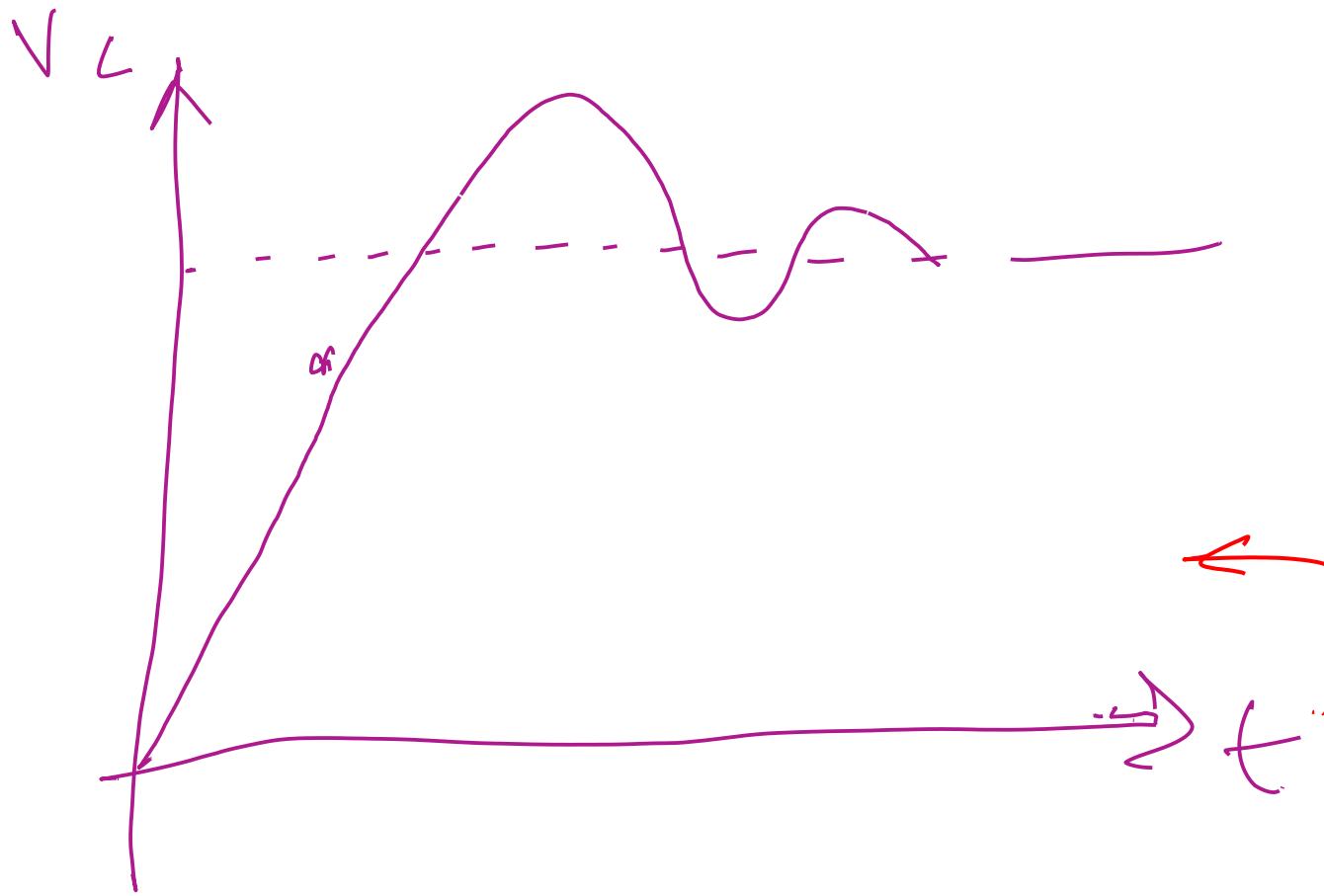
If, $\left(\frac{R}{2L}\right) < \frac{1}{LC}$, then it is called under-damped case. Here the roots are imaginary and unequal

$$\text{So, } V_C(t) = e^{-\alpha t} [A \cos \omega_d t + B \sin \omega_d t]$$

Here,

$$\alpha = -\frac{R}{2L}; \quad \omega_d = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

[see class lecture for the elaborate discussion of derivation of this ~~equation~~ Formula.]

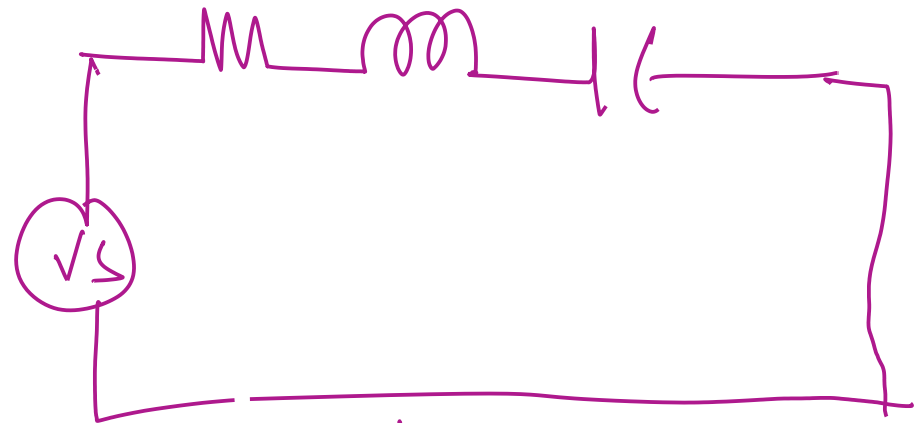


← response of underdamped case.

It is called underdamped case.

at steady state;

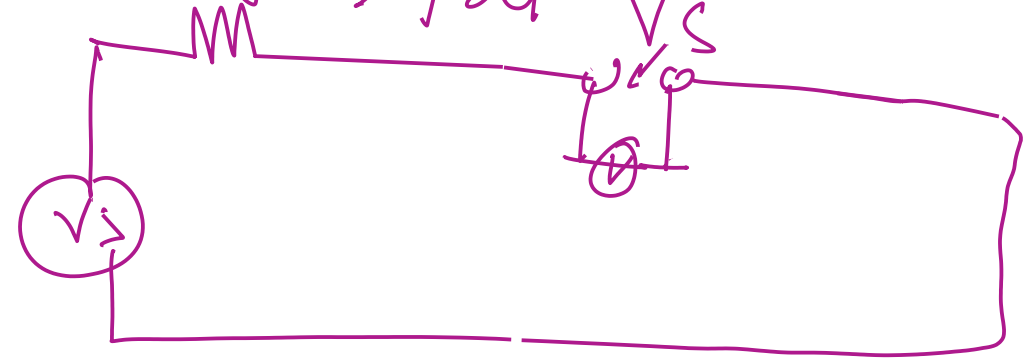
$$V_{SS} = V_S$$



• inductor short

• capacitor open

at steady state i_S



Now. the total response:-

$$V_c(t) = V_{ss} + V_{tr}$$

For overdamped, $V_c(t) = V_{ss} + A_1 e^{s_1 t} + A_2 e^{s_2 t}$

For critical damped, $V_c(t) = V_{ss} + A_1 e^{s t} + A_2 e^{s t}$

For underdamped, $V_c(t) = V_{ss} + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$.

